

BREWSTER FIELD, NE, USAWMO: **725628**Lat: **40.450N**Lon: **99.339W**Elev: **702**StdP: **93.17**Time Zone: **-6.00 (NAC)**Period: **96-19**WBAN: **04935****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.6	-15.1	-22.4	0.5	-16.7	-19.6	0.7	-14.3	13.6	-0.5	12.1	-0.4	4.0	350	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	35.0	22.1	33.0	21.7	31.9	21.8	24.6	30.9	23.7	30.0	22.9	29.2	5.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	19.0	27.9	22.1	18.3	27.4	21.1	17.2	26.5	78.5	30.8	75.0	29.7	71.5	29.4	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	11.2	9.9	DB	-21.8	38.0	1.8	1.6	-23.1	39.1	-24.2	40.0	-25.2	40.9	-26.6	42.1	(4)
(5)			WB	-21.9	26.5	2.1	0.8	-23.4	27.0	-24.6	27.5	-25.8	28.0	-27.3	28.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-2.1	-1.0	4.9	10.1	16.0	21.9	24.5	22.8	18.8	11.4	4.8	-1.4	(6)
(7)		DBStd	10.76	6.41	6.47	6.35	5.34	4.82	3.75	3.06	3.06	4.54	5.26	5.39	5.91	(7)
(8)		HDD10.0	1504	376	310	180	62	7	0	0	0	1	45	169	353	(8)
(9)		HDD18.3	3252	634	542	418	251	104	11	2	3	49	221	406	611	(9)
(10)		CDD10.0	1848	0	2	21	65	193	356	448	397	264	89	13	1	(10)
(11)		CDD18.3	554	0	0	1	3	31	117	191	142	63	6	0	0	(11)
(12)		CDH23.3	5677	0	1	17	81	393	1223	1928	1252	677	99	5	0	(12)
(13)		CDH26.7	2341	0	0	3	21	136	502	888	505	262	26	0	0	(13)
(14)	Wind	WSAvg	4.6	4.4	4.8	5.3	5.8	5.3	4.7	3.7	3.4	4.2	4.6	4.5	4.4	(14)
(15)	Precipitation	PrecAvg	656	14	15	46	54	109	103	98	79	59	40	26	12	(15)
(16)		PrecMax	1178	54	47	168	137	236	246	391	213	251	156	94	38	(16)
(17)		PrecMin	407	0	0	0	0	5	24	6	14	0	0	0	0	(17)
(18)		PrecStd	152	12	13	38	33	56	58	79	44	52	35	25	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.5	20.7	25.3	29.7	33.0	36.0	37.7	36.5	34.7	30.2	23.7	16.2	(19)
(20)			MCWB	7.7	9.9	12.6	14.7	18.1	21.2	22.6	22.1	19.9	17.0	11.8	7.3	(20)
(21)		2%	DB	12.6	15.0	22.2	25.9	30.2	33.7	35.2	33.7	32.3	26.1	19.9	12.2	(21)
(22)			MCWB	5.7	7.1	11.5	14.2	18.5	21.3	23.2	22.6	19.9	15.6	10.6	6.2	(22)
(23)		5%	DB	9.1	11.7	18.7	22.5	27.6	32.0	33.0	31.8	29.9	23.0	16.4	9.0	(23)
(24)			MCWB	4.0	5.9	10.7	13.1	17.3	21.3	22.9	22.7	19.5	14.1	9.0	4.4	(24)
(25)		10%	DB	6.4	7.8	15.3	19.9	25.1	29.7	31.8	29.6	27.4	20.6	13.0	6.6	(25)
(26)			MCWB	2.4	3.7	8.8	12.1	16.6	20.4	22.8	22.1	18.2	13.2	7.6	2.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.9	10.4	15.7	16.9	21.3	24.7	26.1	25.5	23.0	19.4	13.8	9.8	(27)
(28)			MCD B	16.1	17.7	20.4	23.5	28.2	30.7	32.4	31.5	29.7	25.4	18.7	15.5	(28)
(29)		2%	WB	5.9	7.6	13.6	15.5	20.1	23.0	24.9	24.4	21.7	17.4	11.6	6.3	(29)
(30)			MCD B	11.9	14.2	19.1	22.6	26.9	30.1	31.5	29.9	28.0	22.1	17.1	11.6	(30)
(31)		5%	WB	4.4	5.8	11.3	14.0	18.9	22.1	24.0	23.4	20.6	15.7	9.9	4.6	(31)
(32)			MCD B	8.6	11.0	17.2	21.2	25.5	29.3	30.6	29.1	26.8	21.0	16.0	8.9	(32)
(33)		10%	WB	2.8	3.9	9.1	12.7	17.5	21.1	23.3	22.5	19.6	14.0	7.9	2.9	(33)
(34)			MCD B	6.1	7.9	15.8	18.7	23.0	28.0	29.7	28.0	25.5	19.4	13.4	6.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.1	11.1	13.5	13.1	12.3	12.3	12.1	11.7	13.5	13.2	12.6	10.7	(35)
(36)			MCDBR	15.7	16.9	19.3	18.0	15.9	15.1	14.9	14.8	16.3	17.3	17.4	14.7	(36)
(37)			MCWBR	9.9	10.1	10.5	9.0	7.0	6.2	6.3	6.1	6.4	8.3	9.7	9.5	(37)
(38)		5% WB	MCD BR	14.5	15.8	17.0	15.7	13.9	13.4	12.8	12.6	13.5	14.1	16.4	14.0	(38)
(39)			MCWBR	9.5	9.9	10.0	8.5	6.9	6.4	6.4	6.3	6.3	7.9	9.6	9.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.270	0.274	0.300	0.334	0.361	0.377	0.395	0.389	0.353	0.311	0.274	0.265	(40)	
(41)		taud	2.506	2.512	2.466	2.387	2.387	2.381	2.367	2.370	2.424	2.522	2.542	2.519	(41)	
(42)		Ebn at Noon	909	956	958	939	915	897	877	874	887	897	897	888	(42)	
(43)		Edh at Noon	73	85	99	115	118	119	120	116	103	83	70	67	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.29	3.09	4.25	5.32	6.13	7.04	6.90	6.01	4.88	3.50	2.53	1.96	(44)	
(45)		RadStd	0.20	0.24	0.35	0.30	0.38	0.41	0.29	0.32	0.29	0.35	0.17	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page